

Faculty of Computing and Information Technology

BMSE3153 Software Project Management

Assignment Report (202601)

Project Title:	Vehicle Service Appointment Management System
Programme:	RSW
Tutorial Group:	G3
Submission Date:	29/8/2026
Tutor Name	Ms. Faiza

Declaration: I/We declare that this assignment is free from all forms of plagiarism and for all intents and purposes is my/our own properly derived work. I/We have read and complied with all the terms and conditions of Tunku Abdul Rahman University of Management and Technology's plagiarism policy.

No	Student Name	Student ID	Contribution (%)	Signature
1	Alyssa Loh Yi Hui	2511564		<i>alyssa</i>
2	Lee Wei Hang	2511598		<i>hang</i>
3	Ong Yi Jia	2513036		<i>jia</i>
4	Lin Chee Sean	2511602		<i>sean</i>
5	Catherine Lim Yun Yi	2513033		<i>catherine</i>
		Total:	100 %	

Note: The submission date and time will be according to the timestamp recorded in **Tutor's Google Classroom** for Assignment Submission.

Table of Contents

1. Project Background	3
1.1 Introduction	3
1.1.1 Project development team	3
1.1.2 Project client	3
1.1.3 Nature of business and reasons	3
1.2 Project costs estimation	3
1.3 Project objective	3
2. Project Team Structure	3
2.1 Team structure	3
2.2 Stakeholders register	3
2.3 Communication plan	3
3. Project Scope	3
3.1 Scope of the system	3
3.2 Exclusions	3
3.3 Software process model	3
4. Project Schedule	3
4.1 Gantt Chart	3
4.2 Assumptions	3
5. Quality Plan	4
5.1 Quality Attribute 1	4
6. Risk Table	5
6.1 Risk 1	5
7. Ethics and Professional conduct in project management	6
7.1 Engagement strategies to key stakeholders	6
7.2 Reasons to disclose the disadvantages of the chosen model	6
7.3 Corrective actions	6
8. References	7
9. Originality Report	8

1. Project Background

1.1 Introduction

The automotive after-sales service industry in Malaysia is undergoing a rapid shift from traditional walk-in workshops toward digital, platform-based service booking. A leading example of this shift is Tuhu Car Care, which this project takes as its reference application. Tuhu is China's largest independent automotive service platform, backed by Tencent, and has pioneered a "one-stop" digital workshop model that integrates online booking with a nationwide network of physical service outlets. (*CAR CARE, 2026*)

Tuhu officially entered the Malaysian market, bringing a modernised, technology-driven "one-stop car service" concept to Malaysian drivers. Since its debut with two initial outlets, Tuhu has expanded rapidly to eight one-stop digital workshops across Malaysia. Through its mobile application, the platform connects vehicle owners with automotive workshops for services including tyre replacement, engine oil changes, brake maintenance, and general car care. Its core value propositions are transparent pricing, upfront cost breakdowns, and live service tracking through the app, and the current app already allows customers to view service details, pricing, and appointment scheduling, find nearby branches, and manage bookings.

The proposed enhancement addresses these limitations by strengthening the platform's weakest links real-time status accuracy, proactive notification, queue coordination, and customer communication and by expanding it into a complete, end-to-end service management ecosystem. Building on Tuhu's existing booking and tracking foundation, the enhanced system introduces intelligent appointment scheduling with real-time slot availability, a dependable real-time service tracking module that keeps the customer's view synchronised with the workshop's actual job status, a technician and queue management capability for the workshop side, and an automated notification and reminder system that reduces reliance on manual or bot-only communication.

The primary motivation behind this project is therefore to close the gap between what the platform promises and what the customer actually experiences. For customers, the enhancement delivers transparency, reliability, and peace of mind through accurate tracking and timely updates. For workshops, it delivers operational efficiency, better slot and technician utilisation, and reduced miscommunication. For the platform owner, it protects brand reputation and strengthens competitiveness in Malaysia's rapidly digitalising automotive service market. This is a hypothetical academic engagement we are using Tuhu as a real-world client scenario for the assignment, not claiming an actual contract with them.

1.1.1 Project development team

The project is undertaken by TechDrive Solutions, a Malaysian-based software development team specializing in mobile and web-based service management applications. Established with a collective experience of over eight years in the IT industry, the team brings strong expertise across mobile application development, backend and cloud services, database design, UI/UX design, and software quality assurance.

The development team consists of approximately 10 to 20 professionals, including software developers, system analysts, UI/UX designers, database administrators, and quality assurance engineers. This multidisciplinary structure enables effective collaboration throughout different stages of the software development lifecycle, including requirements analysis, system design, development, testing, deployment, and maintenance.

The system analysts are responsible for gathering business requirements from the client, analysing existing vehicle service processes, identifying system limitations, and preparing functional and non-functional requirements. The software developers focus on implementing the core system functionalities, including customer registration, vehicle profile management, service appointment scheduling, workshop availability management, and notification features.

The UI/UX designers are responsible for designing an intuitive and user-friendly interface to ensure customers and workshop staff can easily navigate the platform. Meanwhile, database administrators manage the database architecture to ensure efficient storage, retrieval, and security of important information such as customer details, vehicle records, appointment schedules, and service history. The quality assurance engineers conduct various testing activities, including functional testing, usability testing, integration testing, and performance testing, to ensure the system meets the required quality standards.

Over the years, TechDrive Solutions has gained experience in developing booking and management systems for various industries, including retail, logistics, and service-based businesses. This experience has strengthened the team's ability to design scalable, user-friendly, and efficient appointment-based platforms that support real-time operations and customer management.

The team follows an Agile software development methodology, allowing continuous communication, iterative development, and regular feedback between team members and stakeholders. Through sprint planning, progress reviews, and testing cycles, the team can identify potential issues early and make necessary improvements throughout the project development process.

To support effective project management, TechDrive Solutions utilises modern software development tools and technologies, including Flutter for cross-platform mobile application development, cloud-based backend services for system scalability, database management systems for data organisation, and Git-based version control systems for collaborative development.

The team also applies software quality assurance practices and risk management strategies to ensure successful project delivery. Potential risks such as system performance issues, security vulnerabilities, requirement changes, and scheduling delays are monitored throughout the project lifecycle. By applying structured development practices and effective project coordination, TechDrive Solutions is capable of delivering a reliable vehicle service appointment management system that satisfies both customer and business requirements.

1.1.2 Project client



Figure 1.1.2.1 Logo of Tuhu Car Care

The client for this project is Tuhu Malaysia, a division of Tuhu Car Inc., one of the largest automotive service platforms in China. By combining digital convenience with expert vehicle maintenance services, Tuhu Malaysia entered the Malaysian market to provide a modern, technologically advanced automobile service experience. Through its mobile application and network of authorised service centres, Tuhu Malaysia aims to simplify the process of booking car maintenance services while improving the overall customer experience.

Through a single digital platform, Tuhu Malaysia's online-to-offline (O2O) business model connects car owners with local workshops. The Tuhu mobile application allows users to look for car services, compare prices, purchase tires and vehicle maintenance packages, schedule appointments, find nearby branches, and keep track of their service reservations. The platform offers a wide range of services, including tyre replacement, engine oil changes, battery replacement, brake servicing, wheel alignment, air-conditioning maintenance, and general vehicle inspection. By integrating online booking with physical workshops, Tuhu Malaysia provides customers with a convenient and efficient way to manage their vehicle maintenance needs.

As the project's client, Tuhu Malaysia is committed to using digital technologies to improve customer satisfaction and operational effectiveness. The organization continuously seeks to improve its mobile application by introducing new features that streamline appointment scheduling, reduce waiting times, improve communication between customers and workshops, and provide greater transparency throughout the service process. These improvements create persistent loyalty to customers and increase customer trust.

Additionally, Tuhu Malaysia maintains close relationships with its network of partner workshops to guarantee consistent service quality and standard operating procedures throughout all of its branches. By implementing digital technologies, the organization is able to streamline workshop operations, effectively

schedule service appointments, and provide clients a more reliable and smoother automotive service experience.

TechDrive Solutions has been selected as the software development team for this project in order to provide an improved Vehicle Service Appointment Management System for Tuhu Malaysia. By offering more effective appointment scheduling, real-time service tracking, intelligent scheduling, and improved communication features to better assist clients and workshop personnel, the proposed system will concentrate on enhancing the current application.

1.1.3 Nature of business and reasons

Tuhu Car Care Malaysia operates in the automotive after-sales service industry. Its core business is to provide an integrated online-to-offline platform where vehicle owners can check service prices, book appointments, and track their car servicing digitally, while its network of one-stop workshops carries out the physical service (*Hannani Juhari, 2025*), (*Palaci et al., 2016*). The platform's central promise to customers is transparency, upfront pricing, and live service tracking through the app (*Syahrul, 2025*). The client wishes to undertake this enhancement project because, despite this strong digital-first promise, real operational gaps have emerged between what the app promises and what customers actually experience.

The most significant of these gaps is unreliable order and service status tracking, where the information shown in the app does not always reflect the true status of a customer's order or service. This problem is contributed to by the fact that the in-app status is not synchronised in real time with the workshop's and warehouse's actual data, so the customer's view and the workshop's view can diverge; in one documented case, a customer was shown that an item had arrived when it had not (*App Store, 2026*). The consequence is that customers lose trust in the information displayed in the app, are forced to travel to the workshop or chase updates themselves, and may wait a long time on-site for a service or part that is not actually ready and directly undermining the platform's core value proposition of transparency.

Closely related is a weak proactive notification mechanism, which leaves customers uninformed for long periods after booking or paying. This is contributed to by the absence of an automated notification engine that pushes timely updates at each stage of the order lifecycle; in a real case, a customer waited more than a week after full payment without receiving any update from the app. As a consequence, customers experience uncertainty, perceive the service as unreliable, and are more likely to abandon the platform or leave negative public reviews, which in turn damages the brand's reputation and deters prospective users.

A further gap lies in ineffective customer support and communication, where customers who attempt to resolve issues are unable to obtain meaningful help. This is contributed to by an over-reliance on automated bot replies with no smooth escalation to a human agent, together with communication that is fragmented across channels such as the in-app chat and WhatsApp that are not integrated with the customer's actual order record. In the documented case, the bot replies failed to resolve the issue, and workshop staff contacted separately gave information without properly verifying the item's status. The consequence is that problems remain unresolved, customer frustration escalates, staff provide inconsistent answers, and some customers ultimately decide to stop using the app and workshop altogether, representing a direct loss of customers.

Finally, the platform is affected by manual and uncoordinated workshop scheduling, in which appointment slots and technician workloads are not intelligently managed. This is contributed to by the handling of slot availability and technician assignment through manual processes rather than an automated queue-and-capacity system, with no synchronisation between customer bookings and real workshop capacity. As a result, customers who booked in advance may still wait on-site, technicians and service bays are used inefficiently, and the overall experience contradicts the app's own "skip the line" positioning (*Syahrul, 2025*).

Collectively, these problems share a common root cause the platform's supporting processes for status synchronisation, notification, communication, and scheduling have not kept pace with its customer-facing promises. The consequences are both measurable and serious, ranging from loss of customer trust and wasted customer time to negative reviews, reputational damage, and outright customer churn. By enhancing the system to deliver reliable real-time tracking, automated notifications, integrated communication, and intelligent scheduling, the client can close the gap between promise and experience, protect its brand, and strengthen its competitive position in Malaysia's fast-growing digital automotive service market.

1.2 Project costs estimation

This section presents the estimated cost of developing the enhanced Vehicle Service Appointment Management System for Tuhu Car Care Malaysia. The project is planned over a duration of four (4) months with a project team of five (5) members. The costs are organised into four main categories: hardware requirements, software requirements, staff salaries and benefits, and operational costs. An overall summary is presented first, followed by a detailed breakdown of each category.

Category	Month 1	Month 2	Month 3	Month 4	Total (RM)
1. Staff Salaries and Benefits					
Project Manager	5500	5500	5500	5500	22000
Development Team (4 members)	22000	22000	22000	22000	88000
Meal and Transport Allowance	1500	1500	1500	1500	6000
2. Hardware					
Development and test devices	12255	-	-	-	12255
3. Software					
Cloud, APIs and dev licenses	3268	788	788	788	5632
4. Operational Costs					
Internet, electricity, misc	650	650	650	650	2600
Total Estimated Cost (RM)					136,487

Table 1.2.1 Overall Project Costs Estimation

The enhancement project is planned with a clear and practical budget across four months, covering staff salaries, hardware, software, and operational costs. Staff salaries form the largest portion of the budget, which is expected for a software project because it is labour-intensive rather than material-intensive. Hardware, software and operational costs are comparatively lower because the system is a mobile and web application that relies on cloud services rather than physical infrastructure. The total estimated cost for the entire project is RM 136,487. Each category is explained in detail in the following sub-sections.

1.2.1 Hardware Requirements

As the enhanced system is a mobile and web-based application, the hardware required is limited to development workstations and testing devices needed by the project team during the four-month development period. No specialised on-site hardware is required, since the application runs on customers' and workshop staff's existing smartphones and computers. The detailed hardware costs are shown in Table 1.2.2.

Hardware	Model / Specification	Quantity	Unit Cost (RM)	Total (RM)
Development Laptop	Acer Aspire Lite 14 (16GB RAM)	2	2499	4998
Designer Workstation	Apple MacBook Air M2 13"	1	4499	44990
Android Test Device	Samsung Galaxy A16	1	799	799
iOS Test Device	Apple iPhone SE (3rd Gen)	1	1749	1749
Networking	TP-Link Wi-Fi 6 Router	1	210	210
Total Hardware Cost (RM)				12,255.00

Table 1.2.2 Hardware Requirements Costs

Two development laptops are provided for the software developers to build the front-end and back-end of the system, while a MacBook Air is allocated to the UI/UX designer for design work and iOS-related testing. To ensure the application functions correctly across platforms, one Android device and one iOS device are used for real-device testing of the appointment booking, tracking and notification features. A Wi-Fi 6 router provides stable internet connectivity for the development team. The total hardware cost is RM 12,255.00, incurred as a one-off expense in the first month of the project.

1.2.2 Software Requirements

Software components such as cloud hosting, third-party APIs and the payment gateway are budgeted based on the four-month project duration, as they support development, integration and testing throughout the project. Development-related licenses such as design and productivity tools are also calculated over the same period. The detailed software costs are shown in Table 1.2.3.

Software	Name / Plan	Basis	Total (RM)	Justification
Cloud Backend and Database	Supabase Pro	RM 110/month $\times$ 4	440	Hosting, database, authentication and storage for the app.
Maps and Location	Google Maps Platform	RM 300/month $\times$ 4	1200	Workshop and branch location, distance and mapping.
Payment Gateway	iPay88 / Stripe integration	One-off setup	1500	Secure FPX, card and e-wallet payment processing.
Notification Service	Firebase Cloud Messaging + SMS	RM 197/month $\times$ 4	788	Push notifications, reminders and status updates.
UI/UX Design Tool	Figma Professional (3 seats)	RM 65/month $\times$ 4	260	Interface design and prototyping.
Productivity Suite	Google Workspace (5 users)	RM 172/month $\times$ 4	688	Team documentation and communication.
Dev Tools and Store Fees	Play Store + Apple Developer	One-off / annual	756	App publishing and code-signing accounts.
Total Software Cost (RM)			5632.00	

Table 1.2.3 Software Requirements Costs

Supabase is selected as the cloud backend because it provides database, authentication and storage services on an affordable subscription basis, avoiding the cost of purchasing physical servers. The Google Maps Platform powers the workshop-location and branch-mapping features, while the payment gateway integration enables secure cashless transactions through FPX, cards and e-wallets. Firebase Cloud Messaging together with an SMS service supports the automated notification and reminder module is one of the key enhancements of this project. Figma, Google Workspace and the app-store developer accounts support design, collaboration and publishing. The total software cost is RM 5,632.00 over the four-month duration.

1.2.3 Staff Salaries and Benefits

Staff salaries make up the largest part of the project budget. The team consists of five members, each taking a defined role in the development of the system, plus a meal and transport allowance provided to all members. Salaries are estimated based on typical Malaysian market rates for each role over the four-month project. The detailed salary breakdown is shown in Table 1.2.4.

Role	No.	Monthly Cost (each)	Duration	Total (RM)
Project Manager	1	5,500 / month	4 months	22000
Lead Developer	1	5,000 / month	4 months	20000
Software Developer	2	4,500 / month	4 months	36000
UI/UX Designer	1	4,000 / month	4 months	16000
QA Engineer / Tester	1	4,000 / month	4 months	16000
Meal and Transport Allowance	5	300 / month	4 months	6000
Total Staff Salaries and Benefits (RM)				116,000

Table 1.2.4 Staff Salaries and Benefits

The Project Manager (RM 5,500/month) oversees planning, monitoring, risk management and communication with the client. The Lead Developer (RM 5,000/month) is responsible for system architecture and leads the development of the core tracking and scheduling modules, supported by two Software Developers (RM 4,500/month each) who build the remaining modules. The UI/UX Designer (RM 4,000/month) handles interface design and user experience, while the QA Engineer (RM 4,000/month) prepares test cases and performs functional and usability testing. A meal and transport allowance of RM 300 per member per month is provided as a staff benefit. The total staff cost over four months is RM 116,000.

1.2.4 Operational Costs

Operational costs cover the day-to-day expenses required to support development, integration and testing activities during the four-month project duration. These include internet connectivity, electricity, and miscellaneous consumables. The detailed operational costs are shown in Table 1.2.5.

Category	Monthly Cost (RM)	Duration	Total (RM)
Internet / Network Bill	300 / month	4 months	1200
Electricity	250 / month	4 months	1000
Miscellaneous (printing, consumables)	100 / month	4 months	400
Total Operational Cost (RM)			2,600

Table 1.2.5 Operational Costs

A stable internet connection (RM 300/month) is essential for cloud-based development, API integration and team collaboration, while electricity (RM 250/month) powers the development workstations and testing equipment. A small miscellaneous allowance (RM 100/month) covers printing, documentation and other minor consumables. The total operational cost over the four-month period is RM 2,600.

Cost Category	Total (RM)
Hardware Requirements (Table 1.2.2)	12,255.00
Software Requirements (Table 1.2.3)	5,632.00
Staff Salaries and Benefits (Table 1.2.4)	116,000.00
Operational Costs (Table 1.2.5)	2,600.00
Grand Total Estimated Project Cost (RM)	136,487.00

Combining all four categories, the grand total estimated cost for the Vehicle Service Appointment Management System Enhancement project is RM 136,487.00 over the four-month duration. This budget supports a complete and well-prepared development process that delivers a high-quality enhanced system tailored to Tuhu Car Care Malaysia's operational and customer-service needs.

1.3 Benefits Estimation

Beyond the development cost, the enhanced system is expected to generate measurable financial benefits for Tuhu Car Care Malaysia over a five-year period. These benefits arise from recovering revenue currently lost to missed appointments, attracting and retaining more customers through a better booking experience, and reducing customer churn and manual support effort. The three main benefit categories are estimated below and later used in the NPV and ROI analysis in Section 1.4.

1.3.1 Increased Revenue from Reduced Missed Appointments

A major weakness of the current platform is the high rate of missed or no-show appointments caused by the lack of automated reminders and reliable tracking. By introducing automated reminders and dependable real-time tracking, more booked appointments are completed, recovering service revenue that would otherwise be lost. The estimated recovered revenue is shown in Table 1.3.1.

Item	Year 1	Year 2	Year 3	Year 4	Year 5
Extra appointments completed	600	720	840	960	1080
Revenue recovered (RM)	60,000	72,000	84,000	96,000	108,000

Table 1.3.1 Revenue Recovered from Reduced Missed Appointments (5 Years)

Assuming an average service value of approximately RM 100 per recovered appointment, reducing missed appointments translates directly into recovered revenue that grows each year as adoption of the reminder feature increases.

1.3.2 Increased Revenue from Higher Booking Volume

An improved user experience, including live tracking, smoother scheduling and timely notifications, attracts new customers and encourages existing customers to book more frequently. This increases the overall booking volume and the associated service revenue, as estimated in Table 1.3.2.

Item	Year 1	Year 2	Year 3	Year 4	Year 5
Additional bookings	800	1000	1200	1400	1600
Revenue earned (RM)	80,000	100,000	120,000	140,000	160,000

Table 1.3.2 : Revenue from Increased Booking Volume (5 Years)

As customer trust and satisfaction improve, the platform captures a larger share of repeat and referral bookings, producing a steadily increasing revenue stream across the five-year period.

1.3.3 Reduced Customer Churn and Support Cost

The current reliance on bot-only replies and fragmented communication leads to unresolved complaints and customer loss. Integrated communication and reliable status updates reduce the volume of complaints and the manual effort spent resolving them, while retaining customers who would otherwise leave. The estimated savings are shown in Table 1.3.3.

Item	Year 1	Year 2	Year 3	Year 4	Year 5
Customers retained / cases avoided	200	240	280	320	360
Savings earned (RM)	20,000	24,000	28,000	32,000	36,000

Table 1.3.3 Savings from Reduced Churn and Support Cost (5 Years)

By lowering churn and reducing the manual support workload, the enhancement protects existing revenue and reduces operating cost, contributing further to the overall return on investment.

Benefit Category	Year 1	Year 2	Year 3	Year 4	Year 5
Reduced missed appointments	60,000	72,000	84,000	96,000	108,000
Higher booking volume	80,000	100,000	120,000	140,000	160,000
Reduced churn and support cost	20,000	24,000	28,000	32,000	36,000
Total Benefits (RM)	160,000	196,000	232,000	268,000	304,000

Table 1.3.4 Total Estimated Annual Benefits (5 Years)

1.4 NPV, ROI and Payback Period

To evaluate whether the project is financially worthwhile, the Net Present Value (NPV), Return on Investment (ROI) and Payback Period are calculated using a discount rate of 10% over a five-year period. Year 0 represents the one-off development cost of the project, while Years 1 to 5 include the annual operating (recurring) costs and the benefits generated after the system is launched. Discount factors are rounded to four decimal places.

Cost	Y0	Y1	Y2	Y3	Y4	Y5
Development cost	136,487	-	-	-	-	-
Annual operating cost	-	15,348	15,348	15,348	15,348	15,348
Total cost (RM)	136,487	15,348	15,348	15,348	15,348	15,348
Discount factor (10%)	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209
Discounted cost (RM)	136,487	13,953	12,684	11,531	10,483	9,530

Table 1.4.1 Discounted Costs

The annual operating cost of RM 15,348 covers recurring cloud hosting, maps, notification services, productivity tools and system maintenance after launch. The total discounted cost over the five-year period is RM 194,668.

Benefit	Y0	Y1	Y2	Y3	Y4	Y5
Total benefits (RM)	-	160,000	196,000	232,000	268,000	304,000
Discount factor (10%)	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209
Discounted benefit (RM)	-	145,456	161,974	174,302	183,044	188,754

Table 1.4.2 Discounted Benefits

The total discounted benefit over the five-year period is RM 853,530.

Measure	Result
Total Discounted Benefits (RM)	853,530
Total Discounted Costs (RM)	194,668
Net Present Value (NPV) = 853,530 – 194,668	RM 658,862
Return on Investment (ROI) = (658,862 / 194,668) × 100%	338.45 %
Payback Period	Year 2

Table 1.4.3 NPV, ROI and Payback Period

At a discount rate of 10%, the project produces a positive Net Present Value of RM 658,862, indicating that the total discounted benefits far exceed the total discounted costs over five years. The Return on Investment of 338.45% shows that the project returns significantly more than it costs, and the Payback Period falls within Year 2, meaning the cumulative discounted benefits recover the initial investment early in the second year of operation. Based on these results, the Vehicle Service Appointment Management System Enhancement project is financially attractive and worth pursuing.

1.3 Project objective

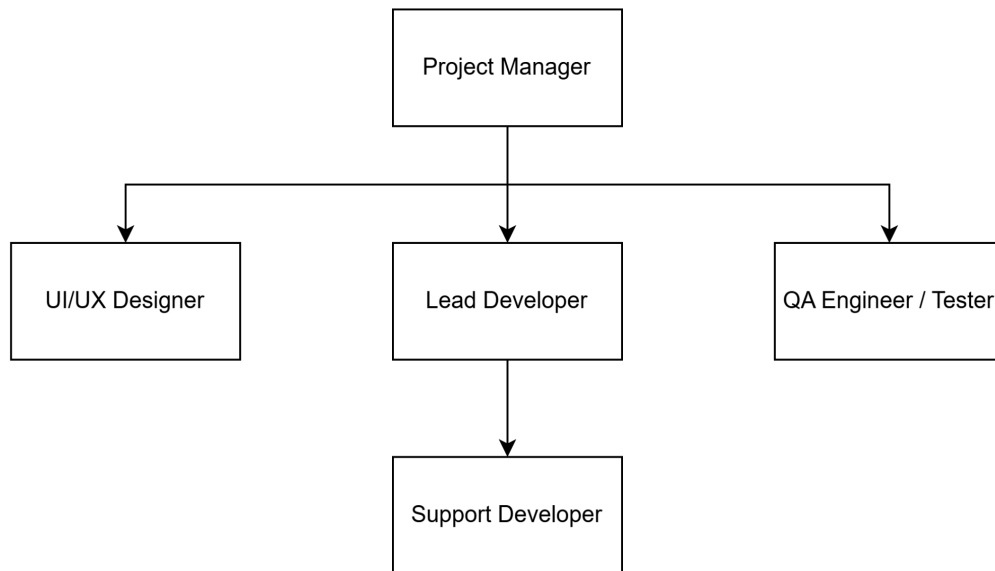
To develop and deploy an enhanced Vehicle Service Appointment Management System for AutoCare Network Sdn. Bhd. within four months that provides real-time service tracking and automated appointment scheduling, thereby reducing missed appointments by at least 30% upon launch.

This objective conforms to the SMART criteria which are specific. It enhanced the system with tracking and scheduling. Next is a measurable 30% reduction in missed appointments. Then, achievable within team capability. Last relevant aligned with client business goals and time-bound in four-month duration.

2. Project Team Structure

2.1 Team structure

The project team of TechDrive Solutions is organised in a functional hierarchy led by a Project Manager, who oversees the design, development, and quality assurance functions. The team structure is illustrated below, followed by the list of members, positions, and roles. In accordance with the assignment requirement, each member also takes turns acting as the weekly project manager to report progress to the tutor.



No	Team Member	Job Position	Role
1	Lee Wei Hang	Project Manager / Developer	Plans schedule, monitors progress, manages risks, and develops the booking & scheduling module.
2	Alyssa Loh Yi Hui	Lead Developer	Designs system architecture and leads development of the service tracking and technician modules.
3	Lin Chee Sean	Support Developer	Develops user account, vehicle profile, and notification modules; assists integration.
4	Ong Yi Jia	UI/UX Designer	Designs interface mock-ups, user flows, and ensures consistent, user-friendly experience.

5		QA Engineer / Tester	Prepares test cases, performs functional and usability testing, and reports defects.
---	--	----------------------	--

2.2 Stakeholders register

Stakeholder	Role	Responsibilities / Expectations	Power	Interest
Tuhu Malaysia Management (Project Sponsor)	Project Sponsor / Client	Approves project scope, budget, timeline and major deliverables. Expects the enhanced system to improve customer satisfaction and operational efficiency	High	High
Project Manager (TechDrive Solutions)	Project Manager	Plans, coordinates, monitors, and controls the project. Manage risks, budget, schedule, and communication with the client	High	High
Development Team (TechDrive Solutions)	Software Development Team	Develops the mobile application, backend services, database, APIs, and system integration according to project requirements	High	High
Quality Assurance (QA) Team	Testing Team	Performs functional integration, usability and performance testing to ensure the system meets quality standards before deployment	Medium	High
Workshop Managers	Business Users	Manage workshop operations, approve appointments, assign technicians, monitor service queues, and provide	High	High
Workshop	Operational Users	Update vehicle service status,	Medium	High

Technicians		record completed work, and use the technician scheduling module		
Customers (Vehicle Owners)	End Users	Book appointments, receive notifications, track vehicle service progress, and provide feedback on user experience	Low	High
System Administrator	IT support	Maintains the production environment, manage user accounts, performs backups, monitor system performance, and ensures system security	Medium	Medium
Third-Party Service (Google Maps, Payment Gateway, Firebase)	External Stakeholders	Provide location services, payment processing and push notification services required by the system	Medium	Medium

Table 2.2 Stakeholder Register

The register reflects five stakeholder groups relevant to this project: internal project stakeholders, including the Project Manager and TechDrive Solutions' development team; the client, Tuhu Malaysia management; business-side users, such as workshop managers and technicians; end users, namely customers; and technical or external support, including the system administrator and third-party API providers. This spread ensures the analysis captures both those who direct the project and those whose day-to-day work or experience the system directly affects.

Tuhu Malaysia management and the Project Manager sit in high power, high interest; the development team, QA, and workshop managers sit in medium-high power, high interest; customers and technicians sit in low-medium power, high interest; and the system administrator and third-party providers sit in medium power, medium interest.

2.3 Communication plan

The communication plan defines how information is shared among stakeholders throughout the project to ensure alignment and transparency.

No	Method	Purpose	Frequency	Audience	Owner
1	Weekly Progress Meeting	Report progress, discuss issues and next steps	Weekly	Project team & tutor	Weekly PM
2	Client Review Meeting	Present milestones and gather client feedback	By-weekly	Client & PM	Project Manager
3	Team Stand-up (WhatsApp / Discord)	Quick daily updates and task coordination	Daily	Project team	Lead Developer
4	Email / Google Classroom	Formal document sharing and approvals	As needed	All stakeholders	Project Manager
5	Sprint Demo	Demonstrate completed features each iteration	End of each sprint	Client & team	Lead Developer

3. Project Scope

3.1 Scope of the system

The scope of the enhanced Vehicle Service Appointment Management System is presented in three levels: the overall system, its modules, and the functions within each module. The system comprises twelve (12) modules covering the customer, workshop, technician, and administrator perspectives. The scope hierarchy is illustrated in Figure 3.1, followed by a description of each module.

No	Module	Functionalities
1	User Account & Profile Management	- Register/login (email, phone, Google OAuth) - Edit profile - Password reset & security - Notification preferences - Account deactivation
2	Vehicle Profile Management	- Add multiple vehicles (plate, brand, model, year) - Per-vehicle service history - Mileage tracking and service-due reminder - Store road tax or insurance expiry - Update or delete records.
3	Service Catalogue Management	- Create & categorise services - Set price ranges - Add descriptions and estimated duration - Enable and disable services - Create service packages
4	Workshop & Branch Management	- Register workshop and profile - Set operating hours and holidays - Map location via Google Maps - Upload photos and certifications - Toggle active status
5	Technician Management	- Create technician accounts - Assign specialisations - Daily job assignment - Availability schedule

		- Performance tracking
6	Appointment Booking & Scheduling	- Browse services and select workshop - Real-time slot calendar - Book, reschedule, cancel - Walk-in vs pre-booking - Booking confirmation with reference number
7	Queue & Slot Management	- Configure slots per time block - Daily queue dashboard - Manual slot blocking - Waitlist for full slots - Auto slot release on cancellation
8	Real-Time Service Tracking	- Live job status (Queued → In Progress → Quality Check → Completed) - Estimated completion time - Technician remarks - Milestone push notifications - Digital inspection checklist
9	Notification & Reminder System	- Booking confirmation - 24-hour & 1-hour reminders - Service status updates - Road tax/insurance expiry reminders - Promotional broadcasts
10	Payment & Invoice Management	- Multiple payment methods (FPX, TNG eWallet, card) - Price quotation - Online payment & receipt - Payment history & invoices - Refund handling
11	Review, Rating & Feedback System	- Rate workshop & technician (1–5 stars) - Submit written review - Workshop reply - Report inappropriate reviews

		- Aggregate rating display
12	Admin Dashboard & Reporting	- User & workshop oversight - Appointment analytics - Revenue & transaction reports - Announcement management - Dispute & complaint handling

Figure 3.1: System Scope (System – Modules – Functions)

3.2 Exclusions

The following items and services are explicitly excluded from the scope of this project:

- Physical automotive servicing and spare-part supply — the system manages appointments and information only; the actual mechanical work and parts inventory remain the responsibility of the partner workshops.
- Hardware procurement and on-site network infrastructure for workshops (e.g., tablets, routers, or POS terminals) are not provided; the client and partner workshops are responsible for their own devices and internet connectivity.

3.3 Software process model

The Incremental (Iterative) model is selected as the software process model for the enhancement of the Tuhu Malaysia Vehicle Service Appointment Management System. According to this model, the system is developed in smaller steps, each of which produces a set of completed and functional modules. After every stage of development, the client receives a functional version of the system rather than waiting until the project is finished.

The proposed system consists of several independent modules, including User Account Management, Vehicle Profile Management, Workshop Management, Appointment Booking, Queue Management, Real-Time Service Tracking, Notification System, Payment Management, Feedback and Rating, and Admin Dashboard. These modules can be divided into multiple development phases according to their priority. For example, the first increment may focus on user registration, vehicle profile management, and appointment booking. Workshop management, queue management, and notification functions can be included in the second increment. The final increments can complete real-time service tracking, payment processing, feedback management, reporting, and system integration.

By using the incremental model, TechDrive Solutions can deliver functional modules incrementally while allowing Tuhu Malaysia to evaluate each increment before the beginning of the next phase. In order to ensure that the final system fulfills the client's operational needs and improves the overall user experience, feedback obtained after each increment can be integrated into the following development. In addition to reducing development risks and enhancing software quality, continuous testing and integration throughout each increment also helps in early identification of errors.

One of the main advantages of the incremental model is that it allows functional modules to be developed and delivered early in the project. For example, the user account management, vehicle profile management, and appointment booking modules can be completed in the first increment before the remaining modules are developed. This makes it possible for Tuhu Malaysia to evaluate completed features, ensure that they satisfy business needs, and provide early feedback. As a result, necessary improvements can be integrated into next increments, reducing the possibility of major changes after the full system is completed.

Another advantage of the incremental model is that each increment is developed, tested, and integrated separately before moving to the next phase. This enables the development team to identify and fix errors earlier, preventing problems from collecting throughout the project. Before integration, elements like appointment scheduling, real-time service tracking, and payment management for the Tuhu Malaysia

system can all be evaluated separately to make sure they work correctly. This approach improves software quality, reduces project risks, and increases the reliability of the final system.

Besides, the disadvantage of the incremental model is that it requires careful planning before development begins. The development team must clearly define the scope, priorities, and dependencies of each module to ensure that all increments can be integrated successfully. If the relationship between modules is not properly planned, integration issues may occur when combining features such as appointment booking, notification services, and payment processing into a complete system. These issues may require additional development time and resources to resolve.

Lastly, another disadvantage is that every increment requires separate planning, development, testing, and client review activities. Since each development phase involves repeated project management and quality assurance processes, the overall workload for TechDrive Solutions is higher than in a single-pass development model. Furthermore, the development effort and management expenses of the project may be increased by multiple testing and review cycles, especially if Tuhu Malaysia requests modifications following each completed increment.

4. Project Schedule

4.1 Gantt Chart

4.2 Assumptions

4.3 Reports

5. Quality Plan

5.1 Quality Attribute

6. Risk Table
6.1 Risk

7. Ethics and Professional conduct in project management

7.1 Engagement strategies to key stakeholders

7.2 Reasons to disclose the disadvantages of the chosen model

7.3 Corrective actions

8. References

- CAR CARE, T. (2026). *Tuhu Car Care*. GooglePlay. https://play.google.com/store/apps/details?id=com.tuhu.android.keepacar&hl=en_us
- Hannani Juhari. (2025, November 22). *TUHU Malaysia – Cara Semak Harga, Bengkel & Tempah Servis*. eCentral.My. <https://ecentral.my/tuhu-malaysia/>
- Palaci, D., Priddy, R., Worms, P., & Ward, K. (2016). *Counting Colours*.
- Syahrul. (2025, November 21). *China's Largest Car Service Chain, TUHU, Officially Enters Malaysia With Two New Outlets*. SpanarBOY Automotive News. <https://www.spanarboy.com/2025/11/21/chinas-largest-car-service-chain-tuhu-officially-enters-malaysia-with-two-new-outlets/>
- App Store. (2026, May 10). App Store. <https://apps.apple.com/my/app/tuhu-car-care-pro-genuine/id6475328621?see-all=reviews&platform=iphone>

9. Originality Report